

now ready. Figure 2 shows how the application will look after the installation.

- Figure 3 gives the hierarchy of the components that we have dragged to the designer.

If you are confused seeing the designer and the components viewers, let me explain these a bit more. Here is the hierarchy that we have placed for our application.

- At the top, we have the title for our application. It's always a good practice to name your application and show it on the screen as well. We have put a label for that and have set its text property to the name of the application.
- Next, we have a horizontal arrangement, which has two labels and their corresponding text boxes.



Figure 4: Button properties

Using the text boxes, you can enter the text. Putting all the components within the horizontal arrangement will keep them aligned horizontally.

- For the *Add* button, we have set the properties in the designer itself (see Figure 4).
- The next horizontal arrangement holds all the actions buttons. As I have explained earlier, keeping them within an arrangement will make the horizontal arrangement their parent and all the components within it, the children.
- For the notifier, keep the default settings as available in the designer. Now we will head towards the block editor to define the behaviour. Let's discuss the actual functionality that we expect from our application.

- The user should be able to add a new book into the existing list by giving the title and the author's name.
- If there is no existing list, the app should create one, and if a list already exists, the new book should be appended to it.
- The user should be able to delete the book details as well, by giving the title and the name of the author.
- The user should be able to view all the stored books.
- The user should be able to clear the whole list by the press of a single button, in case we don't have information regarding the title and author's name.

So let's move on and add these behaviours using the block editor. I hope you remember how to switch from the designer to the block editor. There is a button right above the *Properties* pane to switch between the two.

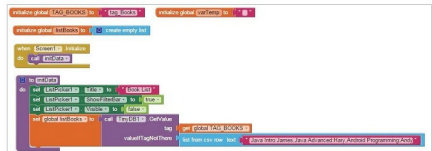


Figure 5: Block Editor Image 1

Block Editor blocks

I have already prepared the blocks for you. All you need to do is drag the relevant blocks from the left side palette and drop them on the viewer. Arrange the blocks in the same way as you see in Figure 5. I will explain what each does and how it is done.

Initialise three variables as follows:

- TAG_BOOKS:** Used for storing the book collection to the database
 - listBooks:** Used to store/manipulate data related to the list of books (title, author, etc)
 - varTemp:** A general-purpose variable
- initData procedure:** On app start-up, it:
- Sets up the *ListPicker* title.
 - Hides the *ListPicker* (we use a button to open it).

- Sets up initial demo data in our database (happens only on app initial install). This procedure attempts to get the data from the database. If data does not exist, then it uses demo data in csv format, splits it using commas (,) and assigns it to our temp.

Screen.Initialize: This is used to check if it's the first time installation. This is done by checking if any database table exists. If not, it will then invoke the *initDatabase* procedure.

sDataEntered: This procedure gets invoked when we try to add a book to the list. It performs validation to ensure data in both fields is entered. It returns 'false' if either the book's name or the author's name is not entered.

isExistingBook: This procedure gets invoked when we try to add or delete a book from the list. It joins the book's title and author in the form of *Title:Author*, and then checks our list to